BRAC UNIVERSITY

Inspiring Excellence

CSE461

Introduction to Robotics Lab

Lab Evaluation Report

Group : 06

Section : 04

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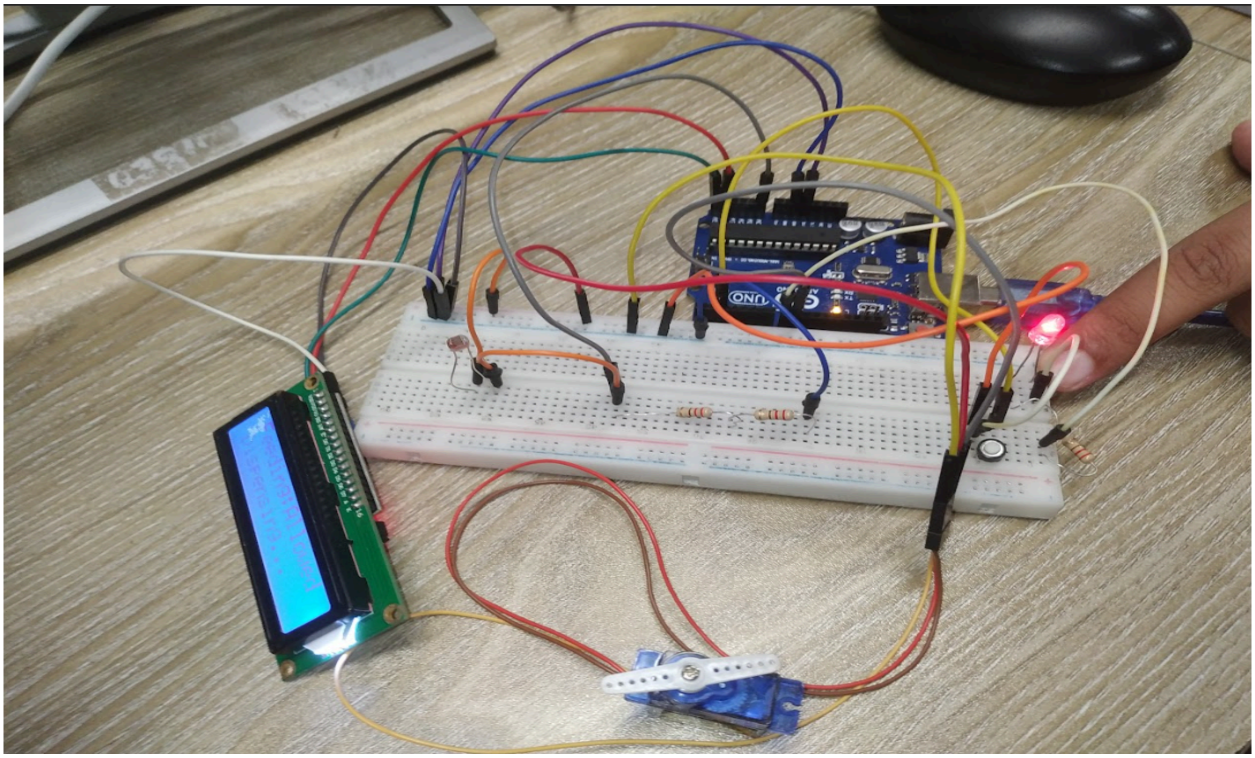
Submitted to - Aabrar Islam and Purba Tahia Hassan

1. **Objectives:** The main objective of this scenario was to design and implement a system with day/night control using basic robotic components called the Smart Pet Feeder. The system is instructed to release food only during the daytime when enough light is detected using an LDR sensor, and a manual trigger button is pressed.
   Additionally, our main goal was learning about the integration of sensors (LDR, button), actuators (servo motor), and output devices (LED and LCD) with a microcontroller to form a simple real-life automated solution with proper condition checking and status indication.

2. **Equipment:**

   - Arduino board
   - Servo Motor
   - LDR
   - LCD Display
   - Push Button
   - LED

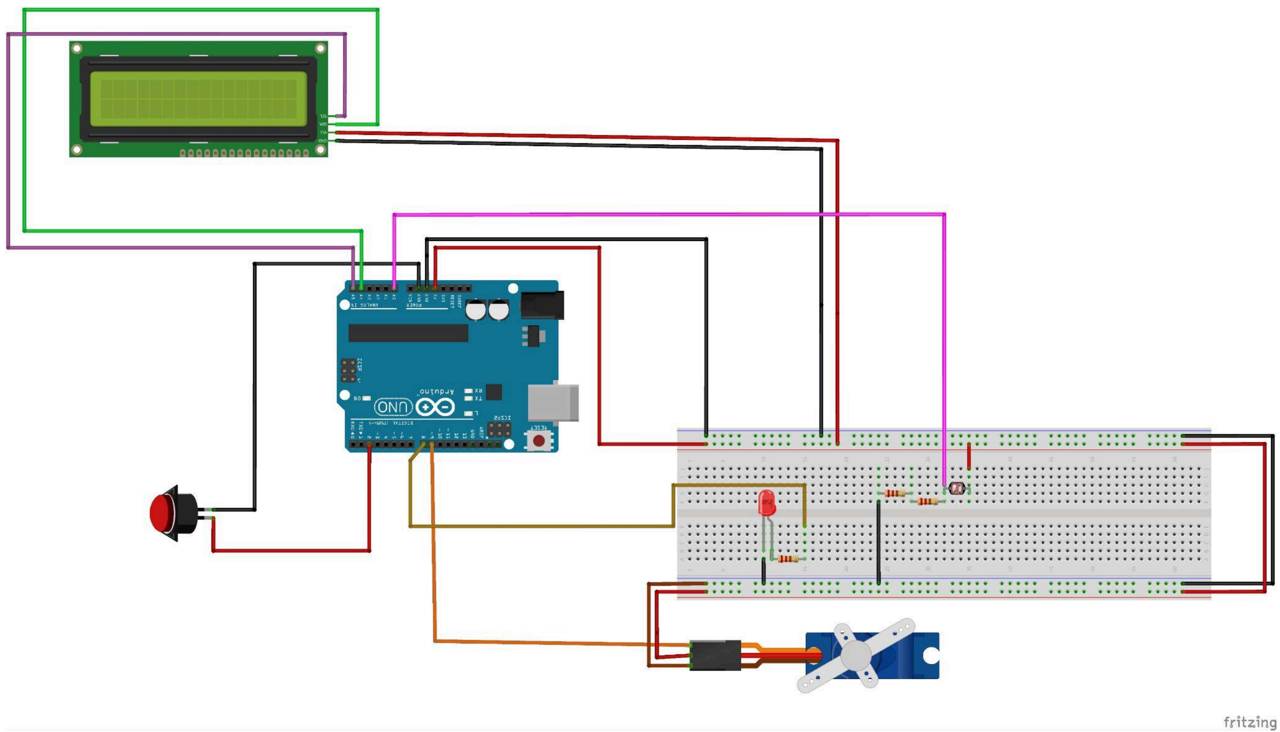
3. **Experimental Setup:**



In this simple experiment, the servo motor was connected to the Audino using the pins VCC, GND and pin D(9). LCD was connected using the pins vcc,gnd [(SDA→A4)(SCL→A5)]. Similarly, LDR was connected to the vcc and the other pin to A0. Likewise, according to the circuit below all the other components were connected.

Circuit Diagram:

- **Lab Approved Circuit: [G-Drive]**
- **Digital copy of the circuit: [G-drive]**



4. **Code:**

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Servo.h>

#define LDR_PIN A0
#define BUTTON_PIN 2
#define LED_PIN 8
#define SERVO_PIN 9

LiquidCrystal_I2C lcd(0x27, 16, 2);
Servo feederServo;

int lightThreshold = 100;
int servoRestPos = 0;
int servoFeedPos = 90;

void setup() {
  Serial.begin(9600);
  Serial.println("Smart Pet Feeder System Started");

  pinMode(LED_PIN, OUTPUT);
  pinMode(BUTTON_PIN, INPUT_PULLUP);

  feederServo.attach(SERVO_PIN);
  feederServo.write(servoRestPos);
  Serial.println("Servo initialized at original position");

  lcd.init();
  lcd.backlight();

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Smart Pet Feeder");

  delay(2000);
  lcd.clear();
}

void loop() {
  int ldrValue = analogRead(LDR_PIN);
  int buttonState = digitalRead(BUTTON_PIN);

  Serial.print("LDR Value: ");
  Serial.println(ldrValue);

  if (ldrValue > lightThreshold) {
    Serial.println("Status: Daytime detected");
    Serial.println("Feeding Status: Allowed");
```

```
    digitalWrite(LED_PIN, HIGH);

    lcd.setCursor(0, 0);
    lcd.print("Feeding:Allowed");

    lcd.setCursor(0, 1);
    lcd.print("LDR:");
    lcd.print(ldrValue);
    lcd.print("       ");

                        if (buttonState == LOW) {
                        Serial.println("Button Pressed");
                        Serial.println("Action: Dispensing food");

                        lcd.clear();
                        lcd.setCursor(0, 0);
                        lcd.print("Feeding:Allowed");
                        lcd.setCursor(0, 1);
                        lcd.print("Dispensing...");

                        feederServo.write(servoFeedPos);
                        Serial.println("Servo rotated to 90 degrees");

                        delay(5000);

                        feederServo.write(servoRestPos);
                        Serial.println("Servo returned to original
position");

                        delay(500);
                        }

                        else {
                        Serial.println("Button not pressed");
                        }


  }


  else {
    Serial.println("Status: Nighttime or Darkness detected");
    Serial.println("Feeding Status: Not Allowed");

    digitalWrite(LED_PIN, LOW);

    lcd.setCursor(0, 0);
    lcd.print("Feeding:Not-Allowed");

    lcd.setCursor(0, 1);
    lcd.print("LDR:");
```

```
    lcd.print(ldrValue);
    lcd.print(" Too Dark");
  }

  Serial.println("--------------------------");
  delay(300);
}
```

5. **Results (Output of the experiment):**

After the successful implementation of the experiment, the smart pet feeder system functioned according to the defined conditions. The ambient light levels were continuously monitored using an LDR sensor to decide whether feeding would be permitted or not. When the intensity of light crossed the predefined value (daytime), the system allowed for feeding and turned the LED ON as an indication of permission to feed.

When the push button is pressed during the daytime, the servo motor rotates by 90 degrees for a short time period (1-2 seconds) to simulate food dispensing and returns to its original position. At this action, the LCD message "Dispensing…" appears.

If the environment was dark (nighttime), the feeding was not allowed, regardless of the button pressing, the LED was still OFF, and the LCD showed "Feeding: Not allowed".

Light Condition	Button Pressed	Servo Action	LED Status	LCD Display
Bright (Day)	**Yes**	**Rotates 90° (Food Dispensed)**	**ON**	**Dispensing…**
Bright (Day)	**No**	**No movement**	**ON**	**Feeding: Allowed**
Dark (Night)	**Yes/No**	**No movement**	**OFF**	**Feeding: Not allowed**

6. Discussions/Answers:

From this lab, we got practical knowledge in designing a condition-based automated system using sensors and actuators. LDR ensured that feeding only occurs during the daytime when the button is pressed, and also prevents nighttime operations. However, there are some issues, like the servo motor sometimes needs to be powered using a different source. Also, as LDR depends on Ambient light so proper threshold detection is tough . Despite all of these limitations, the experiment successfully demonstrates how to use environmental sensing, actuator control and logical conditions and using all of this, we can solve a real-time problem efficiently.

Overall, the smart pet feeder system is a strong example of basic robotics system integration using sensors, actuators, and microcontroller-based control logic and fulfils the lab objectives properly.